

ARUNENDRA TRIPATHI

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EDUCATION

Indian Institute of Information Technology Manipur

Bachelor of Technology (Computer Science and Engineering)

2023 – Present

Manipur, India

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Computer Vision, Operating Systems, DBMS, Computer Networks

INTERNSHIPS

1) Dciphers IT Solutions (P) Ltd.

Greater Noida, India

Software Development Intern

May 2026 – Jul 2026

- Built robust backend services in Go (Golang) for the core monitoring engine, enabling automated discovery of devices across Fortinet firewalls and Wi-Fi networks with real-time monitoring of key device metrics.
- Designed responsive UIs for the Database Access Management System and developed the Forms Management module for the Dciphers Corporate Website.

Internship Project: Network Management System (NMS) | Go, Fiber, PostgreSQL, TimescaleDB, React.js, TypeScript, SNMP, JWT, Docker

- Developed a full-stack NMS with a high-performance Go + Fiber backend implementing REST APIs, JWT authentication, and role/permission-based access control.
- Integrated SNMP for automated device discovery and metric collection, PostgreSQL and TimescaleDB for configuration and time-series data, and a React + TypeScript dashboard for topology visualization, device inventory, and CSV reporting.

2) Indian Institute of Information Technology (IIIT) Vadodara

Vadodara, India

Research Intern – Reversible Data Hiding & Image Security

Jan 2026 – Present

- Developed CNN-based prediction models for Reversible Data Hiding (RDH) using Cross and Dot partitioning techniques, and evaluated CNNP against DP, MEDP, GAP, and BIP through MSE and PSNR-based benchmarking.
- Implemented Histogram Shifting and Prediction Error Expansion for reversible data embedding.

Research Project: Reversible Data Hiding-Based Contrast Enhancement for Medical Images

- Worked on reversible data hiding algorithm for medical image enhancement, integrating superpixel segmentation, multi group histogram stretching, and encrypted image steganography techniques.
- Applied deep learning-based ROI segmentation and implemented evaluation pipelines to measure embedding capacity, image quality, and recovery accuracy.

3) National Institute of Technology (NIT) Warangal

Warangal, India

Research Intern – Machine Learning

Dec 2025 – Jan 2026

- Designed and evaluated machine learning driven optimization techniques for resource allocation and workload management in fog and cloud computing environments.
- Implemented and analyzed Crow Search Algorithm (CSA) variants to improve optimization efficiency across distributed computing scenarios.

Research Project: ScholarSynth (Research Paper Analysis Platform)

[Link](#)

- Built an AI-powered research paper analysis platform using React, TypeScript, Vite, Tailwind CSS, KaTeX, and Google Gemini 2.5 Flash to automate extraction of key research insights from PDF documents.

PROJECTS

1) CodeOrbit : Unified Coding Performance Dashboard

[Link](#)

- Full-stack coding analytics platform using Next.js, React, TypeScript, Node.js, Express.js, MongoDB, and Firebase Authentication to unify coding performance data in one dashboard.
- Integrated statistics from LeetCode, Codeforces, GitHub, CodeChef, and GeeksforGeeks, enabling centralized performance tracking with 365-day activity heatmaps and consistency scoring.
- Integrated LLaMA 3.3 70B (Groq API) for personalized coding recommendations, with automated synchronization workflows and optimized API performance.

2) Worse GPU Allocator : GPU Memory Allocator Simulator

[Link](#)

- Built a dependency-free Python simulator that models GPU memory allocation strategies and their failure modes (fragmentation, oversizing, leaks, CPU fallback) without requiring CUDA hardware.
- Implemented a core WorseGPUAllocator with allocate, free, deterministic seeding, event tracing, and invariant validation for reproducible benchmarking.
- Added CLI demo and benchmark tooling, a full unittest suite, and ruff/mypy checks, documented via an SRS and threat model.

TECHNICAL SKILLS

Languages: Python, C, C++, Go (Golang), JavaScript, TypeScript, SQL

AI/ML: Machine Learning, Computer Vision, Optimization Algorithms, NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow
LLM & GenAI: Prompt Engineering, RAG, Embeddings, Fine-Tuning, AI Agents, LangChain, LangGraph, LlamaIndex, LLM & API Integration (Groq, Gemini)

Web & Backend: React.js, Next.js, Node.js, Express.js, Tailwind CSS, REST API Development, JWT Authentication

Databases & Tools: PostgreSQL, MongoDB, Vector Databases, Git, GitHub, Docker; Data Structures & Algorithms, OOP

ACHIEVEMENTS

- Kaggle Silver Medalist:** Earned a Silver Medal in the “AI Agent Security - Multi-Step Tool Attacks” competition.
- Open-Source Contributions:** Merged PRs to Redis Python client and Red Hat Podman container-lib, earned Hacktoberfest 2024 badges for multiple merged PRs.
- McKinsey & Company Virtual Experience Program (2025):** Solved real-world consulting case problems using structured, data-driven approaches.